

LeapTalent Level 3 End-point Assessment

Data Technician

SAMPLE

Assessment Method:

Scenario Demonstrations with Questioning

90 minutes (for Scenario 1 and 2)

Scenario Demonstration 1 – Data Gathering and Validation

45 minutes

You are employed as a data technician by a policy research institute. The institute is studying how national happiness relates to wealth, health and population, and has asked you to prepare a single, clean data set combining two years of happiness data with population figures, ready for analysis.

You have been provided with the following data sources: **Happiness data 2018.csv**, **Happiness data 2019.xlsx**, **Population data.csv**, and a population **extract** in **Annex A**. Using a suitable data analysis tool of your choice, carry out the following:

Task 1 — Import the data sources

- Import all three data files into your chosen tool.
- Check the column headers and data formats — note that the population file is a World Bank export with several title rows and the data arranged with years down the rows and countries across the columns.

Task 2 — Validate the data

- Check each source for errors or typos in country names or metrics, inconsistencies in column names between the 2018 and 2019 files, and any duplicates or missing values.

Task 3 — Combine the happiness data

- Combine the 2018 and 2019 happiness data into one data set.
- Add a **Year** column to identify which year each row belongs to, and ensure all variables are consistently named and formatted.

Task 4 — Link the population data

- Join the population data to the combined happiness data set using the **country name** as the key, so each country includes its correct population.
- Note that some country names differ between the sources (for example, naming of the Congo, Eswatini/Swaziland, Macedonia) — these must be matched correctly.

Save the prepared data set with a clear name (for example, **Happiness_Population_Combined.xlsx**).

Scenario Demonstration 2 – Data Analysis

45 minutes

Using the cleaned and combined data set from Scenario Demonstration 1, carry out the following analysis and present your findings in a format suitable for senior management.

Task 1 — Compare 2018 and 2019 happiness scores

- For each country, calculate the change in happiness score between 2018 and 2019.
- Identify which countries increased and which decreased.

Task 2 — Average happiness rating

- Calculate the overall average happiness score across all countries, and compare it to the **United Kingdom's** score for both years.

Task 3 — G7 comparison

- Identify the G7 countries (Canada, France, Germany, Italy, Japan, United Kingdom, United States), calculate their average happiness score, and compare it with the global average.

Task 4 — Happiness vs population

- Create a scatter plot of happiness score against population and check for any visible trend or correlation. Consider a logarithmic scale for population.

Task 5 — Explore other variables

- Examine the relationship between happiness score and the other factors (GDP per capita, social support, healthy life expectancy, freedom, generosity, perceptions of corruption) using correlation or scatter plots, and identify which show the strongest relationship.

Annex A – Population Extract (2022)

This extract shows the population figures for a sample of countries, taken from the World Bank population data for the most recent year. Use it to sense-check your linked population values in Scenario 1, Task 4.

Country	Population (2022)
United Kingdom	66,971,411
United States	333,287,557
Germany	84,079,811
Japan	125,124,989
India	1,417,173,173
Finland	5,556,880

MOCK